PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

RescueRoute – Disaster Response and Evacuation Management System

Team Name: **Team #:**	**Crisis Coders** **T-138**
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):*	*Khushi Kalra – 2510380040 2510380040@geu.ac.in* 
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PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

(Brief restatement of your project's main goal. Max 300 words).

RescueRoute is a Disaster Response and Evacuation Management System designed to demonstrate how data structures and algorithms can support emergency coordination. The system models a road network as a graph and uses Breadth-First Search (BFS) to find a path between an emergency location and a hospital. It also includes emergency request prioritization, ambulance management, hospital information, and CSV-based data storage. A priority queue organizes emergency requests according to the implemented severity rules. The project uses simulated locations in Dehradun and provides a console-based prototype. It is an educational project and does not provide live GPS tracking or real-time traffic information.

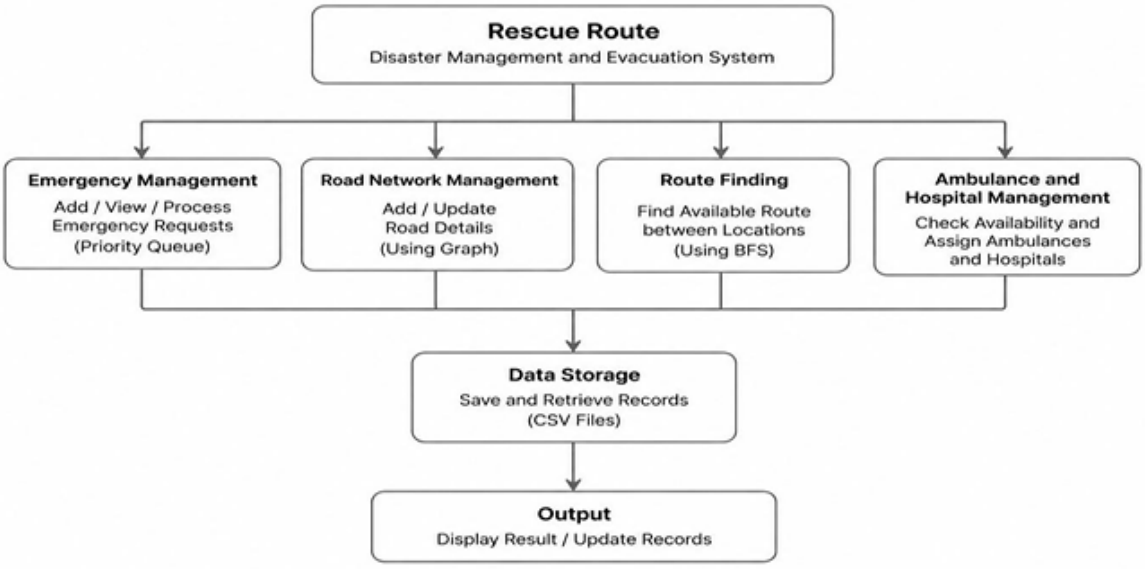
Updated Project Approach and Architecture (2 pts)

(Describe your current approach, including system design, communication protocols, libraries used, etc. Max 300 words).

The project uses C and C++ to implement its data structures and system modules. The road network is represented using an adjacency-list graph. BFS traverses connected locations using a queue and identifies a path when the destination is reachable. Emergency requests are organized using a priority queue according to the system's severity rules. Ambulance records and hospital information are maintained to support simulated emergency assignment.

CSV file handling is used to load and store locations, road connections, emergency requests, ambulances, hospitals, patients, and evacuees. The system is divided into logical components for road-network management, route finding, emergency handling, ambulance assignment, and hospital information.

The prototype is designed to run as a console-based application using local dataset files. BFS finds a path with the fewest edges in an unweighted graph; it does not necessarily identify the shortest geographical distance when road lengths differ. The system currently uses simulated data rather than live map or traffic services.



Tasks Completed (7 pts)

(Describe the main tasks that have been assigned and already completed. Max 250 words).

Task Completed	Team Member
Road-network dataset preparation and graph of locations	Ketan Agarwal
Module structure and BFS route-finding	Khushi Kalra
Emergency request records and priority-queue handling	Kavya Jindal
Ambulance and hospital dataset management	Bhoomi Shukla
Initial CSV dataset preparation and module-level integration	All team members

Challenges/Roadblocks (7 pts)

(Describe the challenges that you have faced or are facing so far and how you plan to solve them. Max 300 words).

The main challenges are making sure that the same location names are used in all the CSV files and the graph putting the data in the format dealing with bad data and making sure that the roads are connected properly. Finding a route can go wrong if the place you want to go to is not connected to where you're starting from or if a needed connection is missing.
These problems are fixed by looking at the CSV headers and the data checking that the location names are correct trying routes, between different places and dealing with situations where no route can be found. Making sure that emergency calls, ambulance details and hospital information work well together also needs testing. The team will keep testing each part of the system and the whole system together to find and fix any problems.

Tasks Pending (7 pts)

(Describe the main tasks that you still need to complete. Max 250 words).

Task Pending	Team Member (to complete the task)
Test BFS routes, disconnected locations, and blocked roads	Khushi Kalra and Ketan Agarwal
Validate emergency priority ordering and request processing	Kavya Jindal
Validate ambulance assignment and hospital capacity handling	Bhoomi Shukla
Complete integration, end-to-end testing, and report updates	All team members

Project Outcome/Deliverables (2 pts)

The main outcome of the project is the console-based system for disaster response and evacuation management. This console-based system for disaster response and evacuation management was made possible by using graph, queue, priority queue, linked list and file operations data structures and algorithms. The code and data set created for this console-based system, for disaster response and evacuation management are stored in the team's GitHub repository.

Progress Overview (2 pts)

The core prototype includes building the graph doing BFS for route finding handling emergency requests and keeping CSV records for ambulances and hospitals. We tested it with sample data. It showed some of the features working. Now we need to test the system with route situations to make sure emergencies get the right priority. Also we should fix how the system handles invalid input. A few more parts need to be added. We'll decide what counts as a result for each part after the final tests. Now the prototype doesn't have live GPS tracking traffic updates or anything, for real-world use.

Codebase Information (2 pts)

(Repository link, branch, and information about important commits.)

GitHub Repository: `https://github.com/KhushiKalra24/RescueRoute`
The repository contains the project's source code and related files. The project uses C and C++ and includes graph-based route finding, emergency handling, ambulance and hospital data management, and CSV file handling.
Repository branch and commit details should be checked against the latest uploaded version before final submission.

Testing and Validation Status (2 pts)

(Provide information about any tests conducted)

Test Type	**Status (Pass/Fail)**	**Notes**
CSV dataset loading	Tested; verify final build	Check headers, record formats, and loaded counts
BFS route finding	Tested; verify final build	Check connected and disconnected locations
Emergency prioritization	Requires final validation	Verify severity ordering and tie-breaking rules
Ambulance and hospital assignment	Requires final validation	Check availability and capacity handling

Deliverables Progress (2 pts)

(Summarize the current status of all key project deliverables mentioned earlier. Indicate whether each deliverable is completed, in progress, or pending.)

Road-network graph and BFS route-finding module: developed; further route validation is pending. Emergency request and priority queue have been implemented in the prototype. The final priority tests for emergency request and priority queue are still pending. Ambulance and hospital data modules are incorporated. CSV dataset has been prepared for locations, roads, emergency requests, ambulances, hospitals, patients and evacuees. Final integration will bring all modules together.